



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE (Advanced)

SCORE - II

01-05-2021

JEE(Main + Advance) : Enthusiast Course (PHASE : SCORE-II)

Test Type : SCORE - II

Test Pattern : JEE (Advanced)

TEST DATE : 01-05-2021

ANSWER KEY

PAPER-1

PHYSICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B	A,C	A,C,D	B,C	A,B,C	B,C,D	C,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	D	C	A	A	C	B	
SEC-3	Q.	1	2	3	4	5		
	A.	6	6	2	8	8		

CHEMISTRY

SEC-1	Q.	1	2	3	4	5	6	7
	A.	A,B,C	B,C,D	A,C,D	A,C	A,B,C,D	A,B	A,B,C,D
SEC-2	Q.	1	2	3	4	5	6	
	A.	A	D	B	D	D	D	
SEC-3	Q.	1	2	3	4	5		
	A.	5	7	3	6	2		

MATHEMATICS

SEC-1	Q.	1	2	3	4	5	6	7
	A.	B,C	A,C	A,B,C	A,C	A,B,C	A,C	A,C
SEC-2	Q.	1	2	3	4	5	6	
	A.	D	C	B	C	A	D	
SEC-3	Q.	1	2	3	4	5		
	A.	9	0	6	4	4		

HINT – SHEET

PHYSICS

SEC-1

2. Ans (A,C)

Inner shell is in unstable equilibrium

$$\Rightarrow U \downarrow \text{ \& } U = \frac{Q^2}{2C}$$

$$\Rightarrow C \uparrow$$

3. Ans (A,C,D)

(A) $dP = Fdt$ (same for both masses)(B) $KE = \frac{P^2}{2m} \Rightarrow KE$ will be different as P is same but m is different(C) Since P is same $\Rightarrow V$ for lighter mass will be more.(D) Since force on both toy will be same $\Rightarrow$ Acceleration of lighter toy will be more

4. Ans (B,C)



5. Ans (A,B,C)

$$(A) \text{ Total distance} = \frac{1000}{5} = 200\text{m}$$

$$w = Fd$$

$$= 600 (200)$$

$$= 120000 \text{ J}$$

$$(B) w = mC \Delta T$$

$$120000 = 1 \times 400 \Delta T$$

$$\Delta T = 300^\circ\text{C}$$

$$(C) M = 0.042 \text{ kg/s}$$

$$= 42 \text{ g/s}$$

$$V = \frac{42\text{g/s}}{5\text{g/m}} = 8.4\text{m/s}$$

6. Ans (B,C,D)

$$m \frac{dv}{dt} = mg - \rho_s V g - 6\pi\eta r v$$

$$\Rightarrow \frac{dv}{dt} = \frac{\rho_s V g - \rho_i V g - 6\pi\eta r v}{\rho_s V g}$$

$$\Rightarrow \frac{dv}{dt} = \underbrace{\left(1 - \frac{\rho_i}{\rho_s}\right)}_A - \underbrace{\frac{6\pi\eta r}{\rho_s V g}}_B v$$

$$\Rightarrow \int_0^v \frac{dv}{A - Bv} = \int_0^t dt$$

$$\Rightarrow -\frac{1}{B} \ln \frac{A - Bv}{A} = t$$

$$\Rightarrow v = \frac{A}{B} (1 - e^{-Bt})$$

$$\therefore \text{Terminal velocity} = \frac{A}{B}$$

$$\therefore 0.9 \frac{A}{B} = \frac{A}{B} (1 - e^{-Bt}) \Rightarrow t \text{ depends on } B \text{ only.}$$

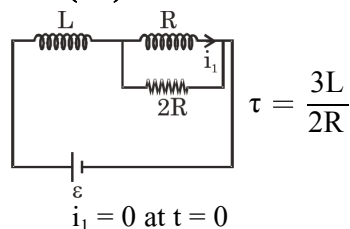
7. Ans (C,D)

Angular momentum about point on ground will remain conserved.

PHYSICS

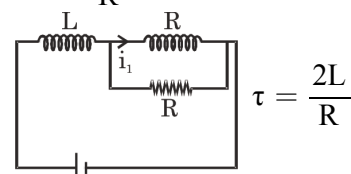
SEC-2

4. Ans (A)



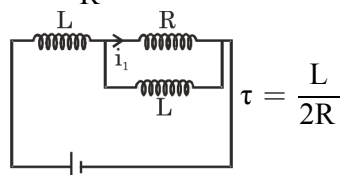
$$i_1 = 0 \text{ at } t = 0$$

$$= \frac{\varepsilon}{R} \text{ at } t \rightarrow \infty$$



$$i_1 = 0 \text{ at } t = 0$$

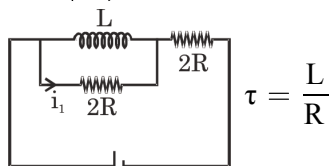
$$i_1 = \frac{\varepsilon}{R} \text{ at } t \rightarrow \infty$$



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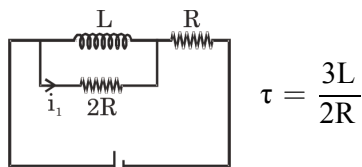
$$= 0 \text{ at } t \rightarrow \infty$$

5. Ans (C)



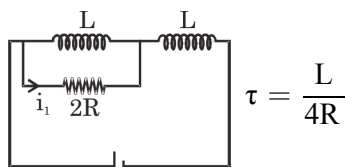
$$i_1 = \frac{\varepsilon}{4R} \text{ at } t = 0$$

$$= 0 \text{ at } t \rightarrow \infty$$



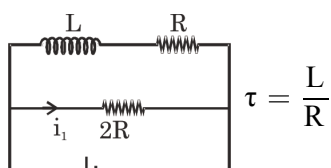
$$i_1 = \frac{\varepsilon}{3R} \text{ at } t = 0$$

$$= 0 \text{ at } t \rightarrow \infty$$



$$i_1 = 0 \text{ at } t = 0$$

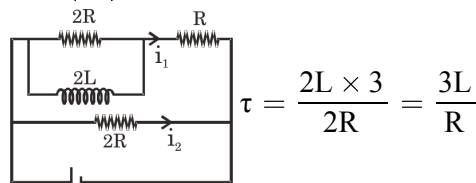
$$= 0 \text{ at } t \rightarrow \infty$$



$$i_1 = \frac{\varepsilon}{2R} \text{ at } t = 0$$

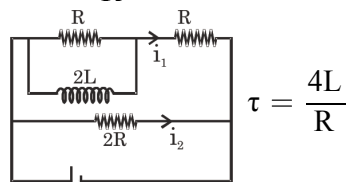
$$= \frac{\varepsilon}{2R} \text{ at } t \rightarrow \infty$$

6. Ans (B)



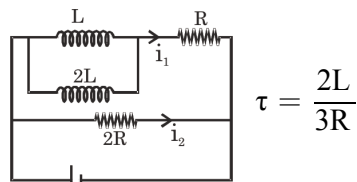
$$i_1 = \frac{\varepsilon}{3R} \text{ at } t = 0$$

$$= \frac{\varepsilon}{R} \text{ at } t \rightarrow \infty$$



$$i_1 = \frac{\varepsilon}{2R} \text{ at } t = 0$$

$$= \frac{\varepsilon}{R} \text{ at } t \rightarrow \infty$$



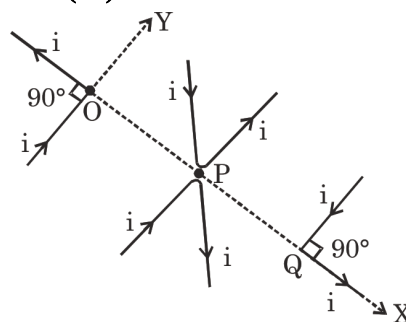
$$i_1 = 0 \text{ at } t = 0$$

$$= \frac{\varepsilon}{R} \text{ at } t \rightarrow \infty$$

PHYSICS

SEC-3

1. Ans (6)



Magnetic field at P is only due to portion AO and BQ and direction of field at P due to both section will be same.

$$B_P = \frac{\mu_0 i}{4\pi \times 2} + \frac{\mu_0 i}{4\pi \times 4} = \frac{3\mu_0 i}{16\pi}$$

2. Ans (6)

$$2d \sin \theta = n\lambda$$

$$\Rightarrow 2 \times 0.6 \times 10^{-9} \times \sin 30^\circ = \lambda$$

$$\lambda = 0.6 \text{ nm}$$

3. Ans (2)

Apply KCL

$$\frac{80}{2R} = \frac{20}{R} + i_2$$

$$i_2 = \frac{20}{R}; i_1 = \frac{80}{2R} = \frac{40}{R}$$

$$\frac{i_1}{i_2} = 2$$

4. Ans (8)

If 100 cm³ liquid has count rate c, the count

rate of x cm³ will be $\left(\frac{C}{100}\right)x$. After 3 half

$$\text{lives, } \left(\frac{C}{100}\right) \times \frac{x}{8} = \frac{C}{10} \Rightarrow x = 80$$

5. Ans (8)

$$\ell_{\text{best}} = \frac{200 \times 5.5}{10.0} = 110$$

$$\text{Now } \frac{\delta \ell}{\ell} = \frac{\delta \ell_1}{\ell_1} + \frac{\delta \ell_2}{\ell_2} + \frac{\delta \ell_3}{\ell_3} = 7.5\%$$

so final answer is $\ell = (110 \pm 8)\text{ft}$

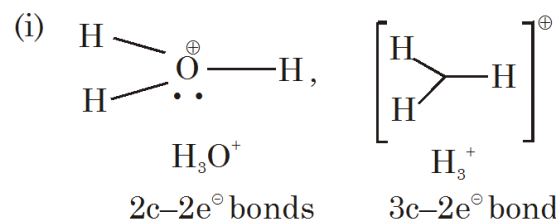
CHEMISTRY

SEC-1

3. Ans (A,C,D)

Liq. matte is placed in converter for self reduction.

6. Ans (A,B)



(ii) In NaHCO₃(s) massive hydrogen bonding is present but KHCO₃(s) exist as dimer.

CHEMISTRY

SEC-3

3. Ans (3)



MATHEMATICS

SEC-1

1. Ans (B,C)

$$\begin{aligned} 3ABA^{-1} + 3A &= 2A^{-1}BA + 2A \\ &= 3A(BA^{-1} + I) = (A^{-1}B + I)2A \\ &= 3A(B + A)A^{-1} = A^{-1}(B + A)2A \end{aligned}$$

Taking mod on both sides

$$3^n|A + B| = 2^n|A + B|$$

$$(3^n - 2^n)|A + B| = 0$$

$$|A + B| = 0$$

$$\text{Let } M = ABA^{-1} - A^{-1}BA$$

$$AM = A^2BA^{-1} - BA \Rightarrow BA = A^2BA^{-1} - AM$$

$$3ABA^{-1} + A = 2A^{-1}BA$$

$$3ABA^{-1} + A = 2A^{-1}(A^2BA^{-1} - AM)$$

$$3ABA^{-1} + A = 2ABA^{-1} - 2M$$

$$-2M = ABA^{-1} + A = A(B + A)A^{-1}$$

$$(-2)^n|M| = |A||A + B||A^{-1}| = 0$$

$$|M| = 0$$

2. Ans (A,C)

$$tx + y = 2t + t^3 \text{ passes through } (13, 12)$$

$$t^3 - 13t - 12 = 0$$

$$\Rightarrow t^2(t + 1) - t(t + 1) - 12(t + 1) = 0$$

$$(t + 1)(t - 4)(t + 3) = 0$$

$$A, B, C = (t^2, 2t)$$

$$A(1, -2); B(16, 8); C(9, -6)$$

$$\text{Centroid} = \left(\frac{26}{3}, 0 \right) \text{ area} = \frac{1}{2} \begin{vmatrix} 1 & -2 & 1 \\ 16 & 8 & 1 \\ 9 & -6 & 1 \end{vmatrix}$$

$$A = 70$$

Area of triangle formed by tangents

$$= \frac{70}{2} = 35$$

3. Ans (A,B,C)

$$|z_1| = 1 \text{ and } |z_2| = 2$$

$$(A) |z_1 + z_2| \leq |z_1| + |z_2| \leq 3$$

$$(B) |2z_1 - z_2| \geq |2|z_1| - |z_2|| \geq 0$$

$$(C) |2z_1 + z_2| \leq 2|z_1| + |z_2| \leq 4$$

$$(D) |2z_1 - 3z_2| \geq |2|z_1| - 3|z_2|| \geq 4$$

4. Ans (A,C)

$$P(W) = \frac{1}{2} P(L) = \frac{1}{6}, P(D) = \frac{1}{3}$$

$P(A \text{ wins}) = (\text{Till } 10^{\text{th}} \text{ match, 1 win and 9 draw})$

$(11^{\text{th}} \text{ match win}) + (\text{Till } 10^{\text{th}} \text{ match}$

$1 \text{ win, 8 draw, 1 lost}) (11^{\text{th}} \text{ match wins})$

$$\frac{10!}{9!1!} \left(\frac{1}{2}\right)^1 \left(\frac{1}{3}\right)^9 \cdot \frac{1}{2} + \frac{10!}{8!2!} \left(\frac{1}{2}\right)^2 \left(\frac{1}{3}\right)^8 \left(\frac{1}{6}\right) \cdot \frac{1}{2}$$

$${}^{10}C_1 \left(\frac{1}{2}\right)^2 \left(\frac{1}{3}\right)^9 + {}^{10}C_2 \left(\frac{1}{2}\right)^2 \left(\frac{1}{3}\right)^9$$

$$(C) P = \frac{{}^{10}C_2}{{}^{10}C_2 + {}^{10}C_1} = \frac{9}{11}$$

5. Ans (A,B,C)

$$f(x) = \{x\} + \{2x\} + \{3x\}$$

$$(1) x \in [0,1) \text{ ND at } x = \frac{1}{3}, \frac{1}{2}, \frac{2}{3} \Rightarrow m = 3$$

$$(2) x \in [0,2) \text{ ND at } x = \frac{1}{2}, \frac{3}{2}, \frac{1}{3}, \frac{2}{3}, \frac{4}{3}, \frac{5}{3}, 1$$

$$\Rightarrow m = 7$$

$$(3) x \in [0,3) \text{ ND at}$$

$$x = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \frac{1}{3}, \frac{2}{3}, \frac{4}{3}, \frac{5}{3}, \frac{7}{3}, \frac{8}{3}, 1, 2$$

$$\Rightarrow m = 11$$

6. Ans (A,C)

$$\text{centre : } \left(\frac{5}{2}, 0\right); y = mx \pm \sqrt{25m^2 - 16}$$

$$\frac{5}{2}m = \pm \sqrt{25m^2 - 16} \Rightarrow \frac{25m^2}{4} = 25m^2 - 16$$

$$\frac{75m^2}{4} = 16 \Rightarrow m^2 - \frac{64}{65} = 0,$$

$$\text{product of slopes of CA, CB is } -\frac{64}{75}$$

circumcentre of ΔABC lies inside the ΔABC

7. Ans (A,C)

$$(a+b) + (b+c) + (a+c) = 8$$

$$\text{let } a+b = \alpha, b+c = \beta, a+c = \gamma$$

$$\Rightarrow \alpha + \beta + \gamma = 8$$

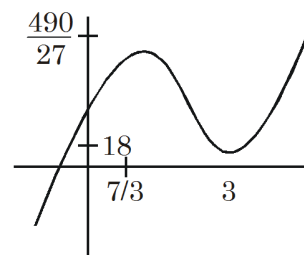
$$\Rightarrow \alpha\beta + \beta\gamma + \gamma\alpha = a^2 + b^2 + c^2 + 3(ab + bc + ac)$$

$$= 21$$

$$\Rightarrow x^3 - 8x^2 + 21x - \lambda = 0$$

$$\text{where } \lambda = (a+b)(b+c)(a+c)$$

these equation has 3 real roots



$$\text{so } f(3)f(7/3) \leq 0$$

$$18 \leq \lambda \leq \frac{490}{27}$$

MATHEMATICS

SEC-2

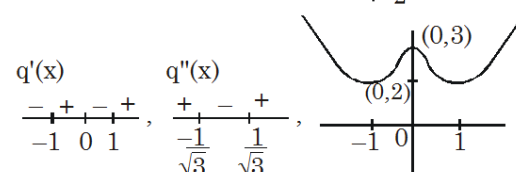
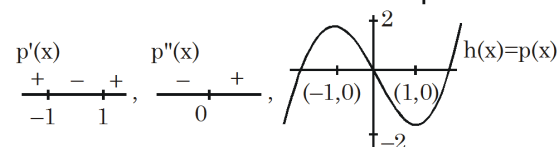
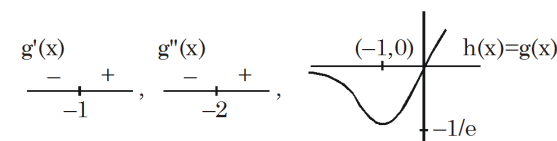
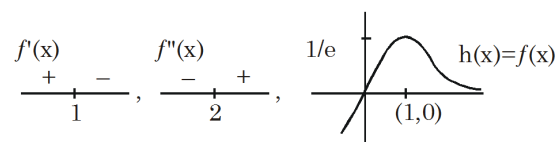
1. Ans (D)

$$f(x) = xe^{-x}, f'(x) = -(x-1)e^{-x}, f''(x) = e^{-x}(x-2)$$

$$g(x) = xe^x, g'(x) = (x+1)e^x, g''(x) = (x+2)e^x$$

$$p(x) = x^3 - 3x, p'(x) = 3x^2 - 3, p''(x) = 6x$$

$$q(x) = x^4 - 2x^2 + 3, q'(x) = 4x(x-1)(x+1), q''(x) = 12x^2 - 4$$



(I), (i)(iv), (R)

(II), (i)(iv), (R,S)

(III), (ii)(iv), (P,Q,R)

(IV), (ii)(iii), (P,S)

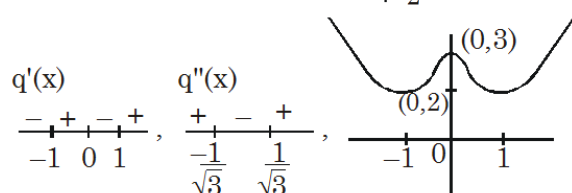
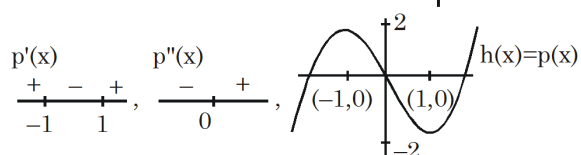
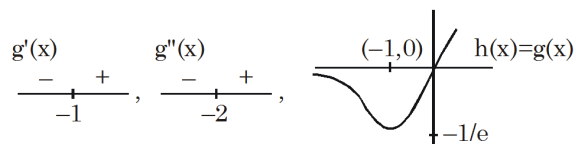
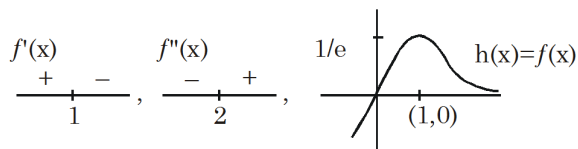
2. Ans (C)

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(I), (i)(iv), (R)

(II), (i)(iv), (R,S)

(III), (ii)(iv), (P,Q,R)

(IV), (ii)(iii), (P,S)

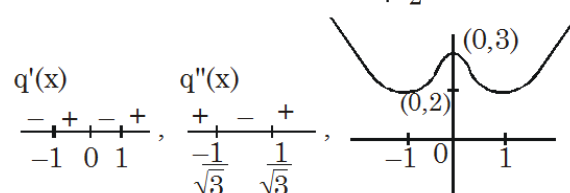
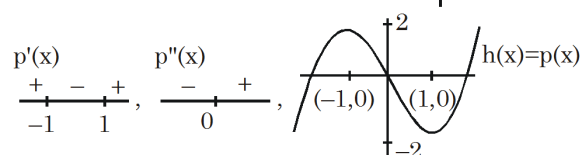
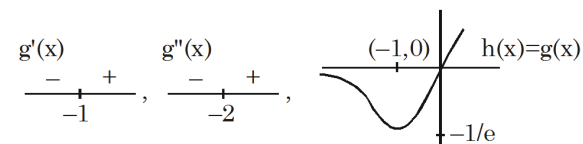
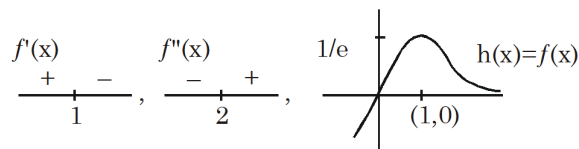
3. Ans (B)

$$f(x) = xe^{-x}, f'(x) = -(x-1)e^{-x}, f''(x) = e^{-x}(x-2)$$

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$$p(x) = x^3 - 3x, p'(x) = 3x^2 - 3, p''(x) = 6x$$

$$q(x) = x^4 - 2x^2 + 3, q'(x) = 4x(x-1)(x+1), q''(x) = 12x^2 - 4$$



(I), (i)(iv), (R)

(II), (i)(iv), (R,S)

(III), (ii)(iv), (P,Q,R)

(IV), (ii)(iii), (P,S)

4. Ans (C)

(I) $ae - \frac{a}{e} = 3$ and $ae + \frac{a}{e} = 5$ we get

$ae = 4$ and $\frac{a}{e} = 1$

$a^2 = 4$ and $e = 2$

(I), (iv), (Q,S)

(II) $x = \frac{e^t + e^{-t}}{2}, y = \frac{e^t - e^{-t}}{3}$

$e^t + e^{-t} = 2x$ and $e^t - e^{-t} = 3y$

squaring and subtracting $4x^2 - 9y^2 = 4$

$e = \frac{\sqrt{13}}{3}$

(II), (iii), (P,R,S)

(III) $P \left(\frac{2a}{\sqrt{3}}, \frac{b}{\sqrt{3}} \right)$

equation of tangent at P is

$\frac{x}{\frac{\sqrt{3}a}{2}} - \frac{y}{\sqrt{3}b} = 1$

Area of triangle $= \frac{1}{2} \times \frac{\sqrt{3}a}{2} \times \sqrt{3}b = 3a^2$

$\Rightarrow \frac{b}{a} = 4, e = \sqrt{17}$

(III), (i), (Q)

(IV) $\frac{a^2x}{6} + \frac{b^2y}{3} = a^2 + b^2$

It passes through (9,0)

$\frac{9a^2}{6} = a^2 + b^2 \Rightarrow \frac{b^2}{a^2} = \frac{1}{2}$

$\Rightarrow e = \sqrt{\frac{3}{2}}$

(IV), (ii), (P,R,S)

If slope of tangent to the hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is m , then $a^2m^2 - b^2 \geq 0$

$\Rightarrow m^2 \geq e^2 - 1$

director circle $x^2 + y^2 = a^2 - b^2$, it exist only

when $a > b \Rightarrow e^2 < 2$

5. Ans (A)

(I) $ae - \frac{a}{e} = 3$ and $ae + \frac{a}{e} = 5$ we get

$ae = 4$ and $\frac{a}{e} = 1$

$a^2 = 4$ and $e = 2$

(I), (iv), (Q,S)

(II) $x = \frac{e^t + e^{-t}}{2}, y = \frac{e^t - e^{-t}}{3}$

$e^t + e^{-t} = 2x$ and $e^t - e^{-t} = 3y$

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(II), (iii), (P,R,S)

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equation of tangent at P is

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Area of triangle $= \frac{1}{2} \times \frac{\sqrt{3}a}{2} \times \sqrt{3}b = 3a^2$

$\Rightarrow \frac{b}{a} = 4, e = \sqrt{17}$

(III), (i), (Q)

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It passes through (9,0)

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(IV), (ii), (P,R,S)

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$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is m , then $a^2m^2 - b^2 \geq 0$

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director circle $x^2 + y^2 = a^2 - b^2$, it exist only

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6. Ans (D)

(I) $ae - \frac{a}{e} = 3$ and $ae + \frac{a}{e} = 5$ we get

$ae = 4$ and $\frac{a}{e} = 1$

$a^2 = 4$ and $e = 2$

(I), (iv), (Q,S)

(II) $x = \frac{e^t + e^{-t}}{2}, y = \frac{e^t - e^{-t}}{3}$

$e^t + e^{-t} = 2x$ and $e^t - e^{-t} = 3y$

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(II), (iii), (P,R,S)

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Area of triangle $= \frac{1}{2} \times \frac{\sqrt{3}a}{2} \times \sqrt{3}b = 3a^2$

$\Rightarrow \frac{b}{a} = 4, e = \sqrt{17}$

(III), (i), (Q)

(IV) $\frac{a^2x}{6} + \frac{b^2y}{3} = a^2 + b^2$

It passes through (9,0)

$\frac{9a^2}{6} = a^2 + b^2 \Rightarrow \frac{b^2}{a^2} = \frac{1}{2}$

$\Rightarrow e = \sqrt{\frac{3}{2}}$

(IV), (ii), (P,R,S)

If slope of tangent to the hyperbola

$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ is m, then $a^2m^2 - b^2 \geq 0$

$\Rightarrow m^2 \geq e^2 - 1$

director circle $x^2 + y^2 = a^2 - b^2$, it exist only

when $a > b \Rightarrow e^2 < 2$

MATHEMATICS

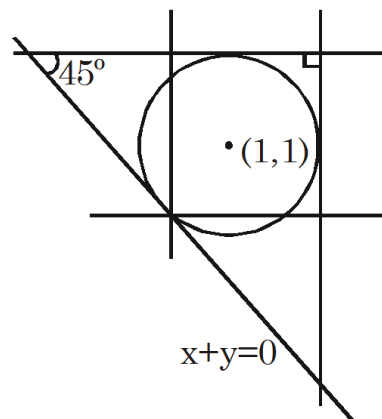
SEC-3

1. Ans (9)

$10! = 2^8 \cdot 3^4 \cdot 5^2 \cdot 7$

Number of divisors $= 9 \times 5 \times 3 \times 2 = 270 = \lambda$

2. Ans (0)



$(h^2 + k^2 - 4)^2 = (h^2 + k^2 - 4h)(h^2 + k^2 - 4k) + 16$

$\Rightarrow (h + k)(h^2 + k^2 - 2h - 2k) = 0$

circle $C_1 : x^2 + y^2 - 2x - 2y = 0$ and

$C_2 : x + y = 0$

circumcentre will be mid point of

hypotenuse in right angle triangle

$(\alpha, \beta) = (0, 0)$

3. Ans (6)

$$D = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & 3 \\ 1 & 2 & a \end{vmatrix} = a$$

unique solution $a \neq 0$

ordered pair $= 6 \times 7$

4. Ans (4)

Differentiating

$$\frac{g^2(x)}{2} = \frac{2xg(x) \int_0^x g(t) dt - \left(\int_0^x g(t) dt \right)^2}{x^2}$$

Let $xg(x) = u$ and $\int_0^x g(t) dt = v$

$$u^2 = 4uv - 2v^2 \Rightarrow u^2 - 4uv + 2v^2 = 0$$

$$u = (2 \pm \sqrt{2})v \Rightarrow xg(x) = (2 \pm \sqrt{2}) \int_0^x g(t) dt$$

let $\int_0^x g(t) dt = f(x) \Rightarrow g(x) = f'(x)$

$$\frac{xf'(x)}{f(x)} = 2 \pm \sqrt{2}$$

$$\Rightarrow \log_e f(x) = (2 \pm \sqrt{2}) \log_e x + c$$

$$f(x) = c_1 x^{2+\sqrt{2}} \text{ or } c_2 x^{2-\sqrt{2}}$$

$$\Rightarrow g(x) = c_3 x^{1+\sqrt{2}} \text{ or } c_4 x^{1-\sqrt{2}}$$

but $g(x)$ is continuous on $[0, \infty)$ so

$$g(x) = c_1 x^{1+\sqrt{2}}, g(1) = 1 \Rightarrow c_1 = 1$$

$$g(x) = x^{1+\sqrt{2}}$$

5. Ans (4)

$$S_n = 1.3 + 2.2^2 + 3.3^3 + 4.2^4 + \dots + \text{upto } n\text{-terms}$$

$$S_n = (1.3 + 3.3^3 + 5.3^5 + \dots) + (2.2^2 + 4.2^4 + 6.2^6 + \dots)$$

$$S_1 = 1.3 + 3.3^3 + 5.3^5 + \dots + \text{upto } 10\text{-terms},$$

$$S_1 = \frac{15}{32} (5.3^{21} + 1)$$

$$S_2 = 2.2^2 + 4.2^4 + 6.2^6 + \dots + \text{upto } 10\text{-terms},$$

$$S_2 = \frac{58.2^{22} + 8}{9}$$

$$S_{20} = S_1 + S_2 = \frac{15}{32} (5.3^{21} + 1) + \left(\frac{58.2^{22} + 8}{9} \right)$$

$$= \frac{75}{32} .3^{21} + \frac{58}{9} .2^{22} + \frac{391}{288}$$